



Roll No: _____

BPHARM

(SEM. VI) THEORY EXAMINATION 2025-26
MEDICINAL CHEMISTRY III – THEORY

TIME: 3 HRS

M.MARKS: 75

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief.

2x10 = 20

- What is Beta-Lactam Ring in Beta-Lactam Antibiotics?
- Draw two chemical structures of Minocycline and Doxycycline
- Briefly describe the therapeutic use of Proguanil
- Define and classify prodrugs with examples.
- Compare Fluoroquinolones and Nalidixic Acid based on spectrum of activity.
- Classify Anti-infective agents.
- Write mode of action and synthesis of Miconazole.
- State the therapeutic use of Folate reductase inhibitors
- Define Hansch analysis.
- Differentiate between rigid docking and flexible docking.

SECTION B

2. Attempt any two of the following:

10 x 2 = 20

- Describe the nomenclature and stereochemistry and SAR of Cephalosporin
- Classify antimalarial agents with examples. Describe the mode of action and synthesis and use of Chloroquine.
- Explain physicochemical parameter used in QSAR.

SECTION C

3. Attempt any five part of the following:

7 x 5 = 35

- Discuss SAR of Quinolones. Draw the synthetic route of Ciprofloxacin.
- Discuss chemistry, classification and SAR of Sulfonamides with suitable examples.
- Describe the steps involved in molecular docking.
- Explain mode of action and therapeutic use of Erythromycin Clarithromycin and Azithromycin.
- Write the synthesis and use of Isoniazid and acyclovir.
- Classify Anti-protozoal agents with suitable examples. Explain the synthesis of Metronidazole
- Define combinatorial chemistry and explain in detail about solid phase synthesis.